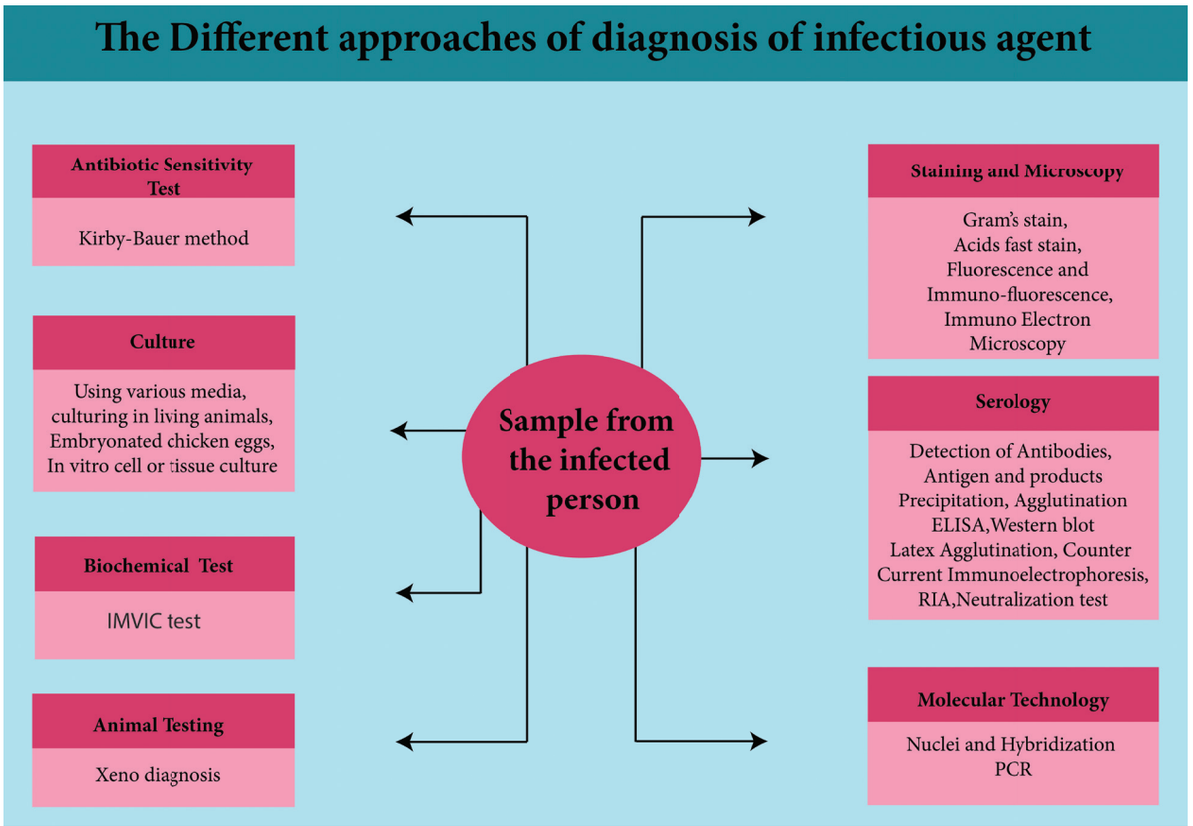


Figure 12.4: Different approaches of diagnosis

Laboratory diagnosis of infectious agents

Direct diagnosis: It is the demonstration of the presence of an infectious agent, antigen or nucleic acids

Indirect diagnosis: It is the demonstration of presence of antibodies to a particular infectious agent, cytopathic effects, haemagglutination, inclusion bodies and neutralization.

The different approaches for diagnosis or identification of infectious agents are shown in Figure 12.4.

12.2 Skin and Wound Infections

The skin, which covers and protects the body, is the body's first line of defense against pathogens. As a physical barrier, it is almost impossible for the pathogens to penetrate it. However, microorganisms can enter through skin breaks that are not readily apparent, and the larval forms of a few parasites can penetrate the intact skin. The skin has up to seven layers (Figure 12.5) of ectodermal tissue and guards the underlying tissues viz; muscles, bones, ligaments and internal organs. Nearly all human skin is covered with hair follicles. Because it interfaces with the environment, skin plays an important role in protecting the body against pathogens and excessive water loss. Its other functions are insulation, temperature regulation, sensation, synthesis of vitamin D, and the protection of vitamin B folates. Severely damaged skin will try to heal by forming scar tissue. This is often discolored and depigmented.

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